

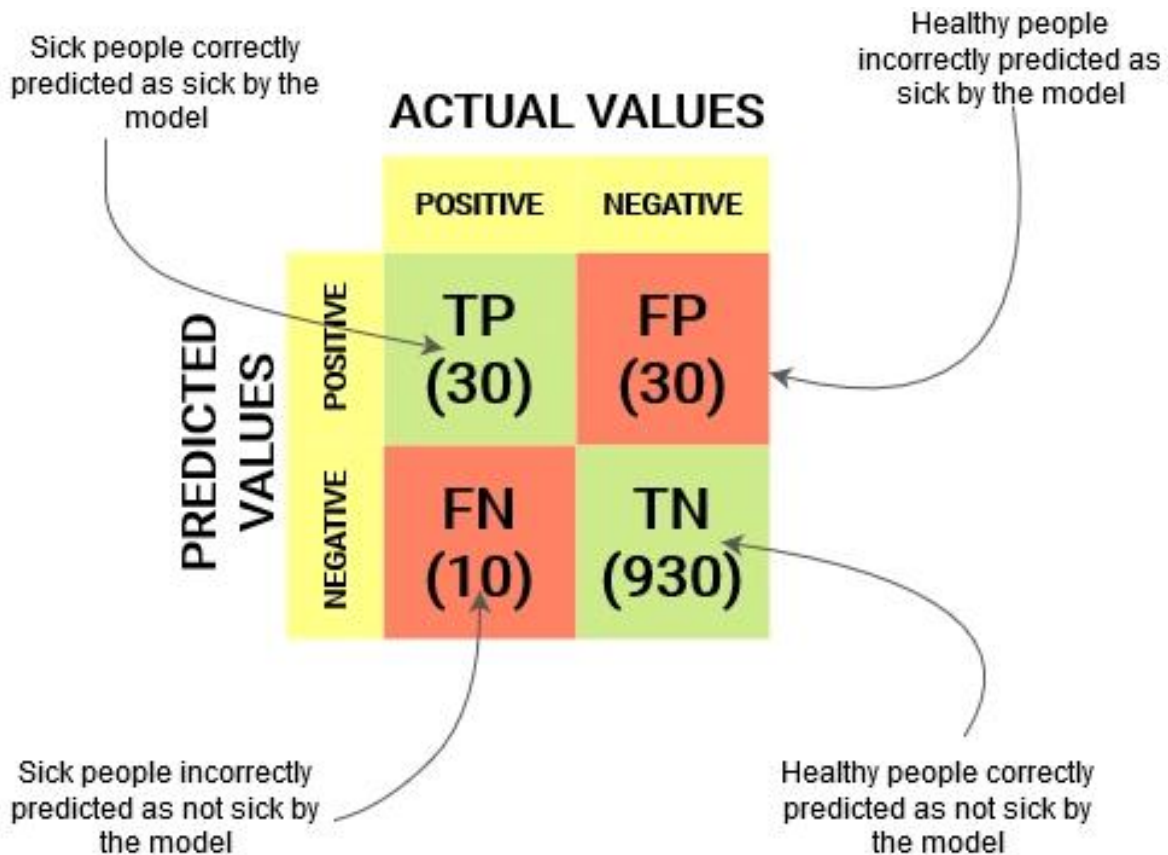
Evaluating Classification Models

*Beyond Accuracy—
Knowing When Your Model
is Truly Successful*

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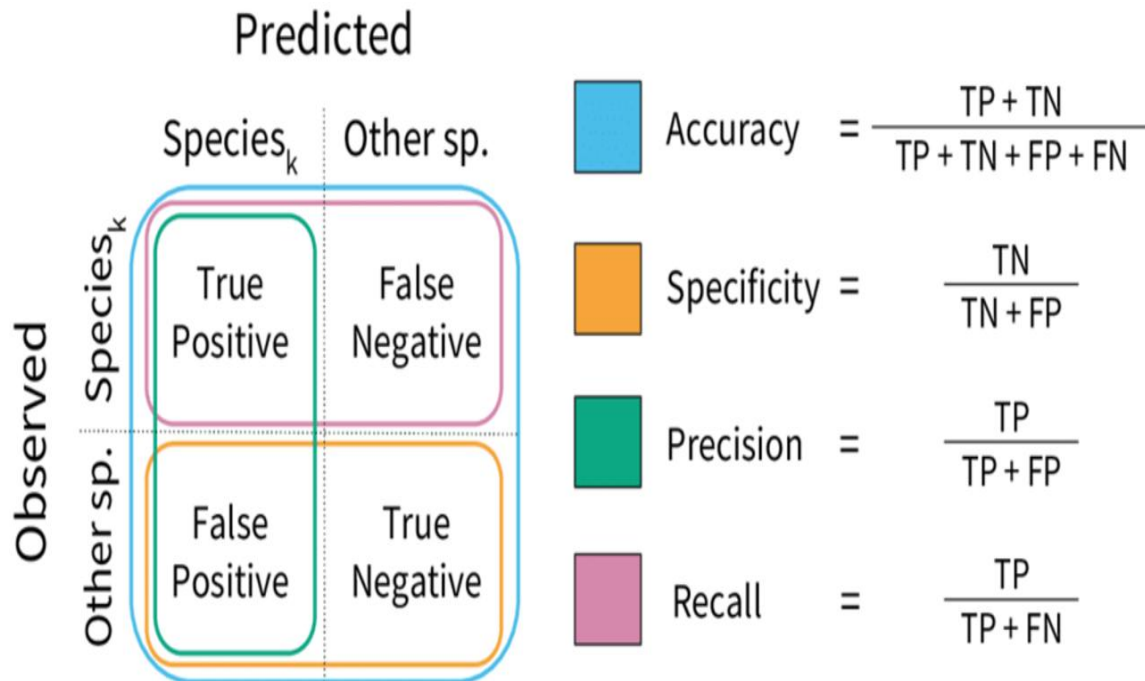
		Predicted			
		+	-		
Actual	+	TP Type I error	FN Type II error	Sensitivity (recall) TP/●	False negative rate FN/●
	-	FP Type I error	TN	False positive rate FP/●	Specificity TN/●
		Precision TP/■	False omission rate FN/■	Accuracy (TP + TN)/(● + ●)	
		FDR FP/■	Negative predictive value TN/■	F_1 score $2TP/(2TP + FP + FN)$	

The Confusion Matrix



- **Definition:** A simple 2x2 grid that shows exactly where the model's predictions match the actual data and where it got "confused".
- **The Four Quadrants:**
 - **True Positive (TP):** Predicted "Yes," and it was "Yes".
 - **True Negative (TN):** Predicted "No," and it was "No".
 - **False Positive (FP):** Predicted "Yes," but it was actually "No" (False Alarm).
 - **False Negative (FN):** Predicted "No," but it was actually "Yes" (Missed Opportunity).

Accuracy vs. Precision vs. Recall



- **Accuracy:** The percentage of total correct guesses. This can be misleading if your dataset is uneven.
- **Precision:** "When the model says 'Yes,' how often is it right?" Important for avoiding false alarms (e.g., Spam filters).
- **Recall:** "Out of all the actual 'Yes' cases, how many did the model find?"
- **The Priority:** In medical screening, **Recall** is more important because missing a sick person (False Negative) is dangerous.

The F1-Score

The F-score

The F-score is the Harmonic mean of Precision and Recall.

$$F = \frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$$

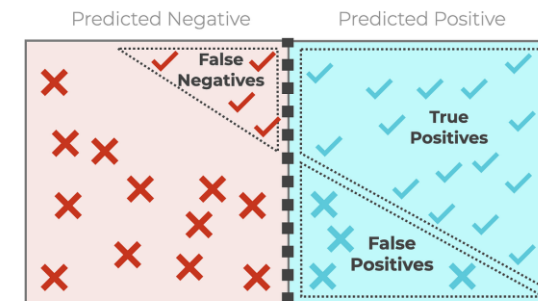
Alternatively

$$F = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Concept:** A combined score that balances both Precision and Recall into a single number.
- **Usage:** Use this when you have an uneven dataset (e.g., 99% of people are healthy, and only 1% are sick).
- **Formula:**

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 SCORE, EXPLAINED



$$\text{F1 Score} = \frac{2 * \left(\begin{array}{c} \text{True} \\ \text{Positives} \end{array} \right)}{2 * \left(\begin{array}{c} \text{True} \\ \text{Positives} \end{array} \right) + \left(\begin{array}{c} \text{False} \\ \text{Positives} \end{array} \right) + \left(\begin{array}{c} \text{False} \\ \text{Negatives} \end{array} \right)}$$